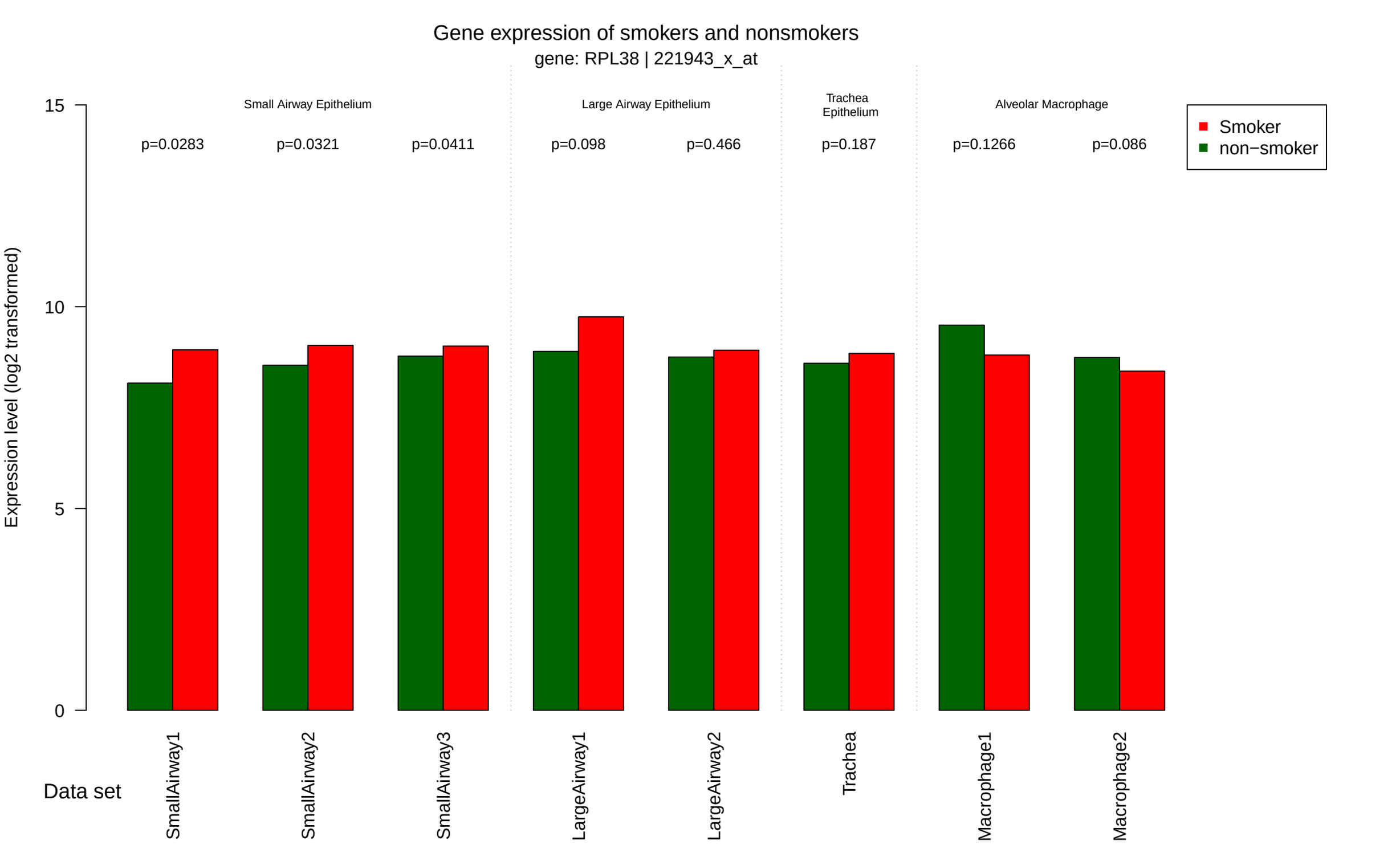
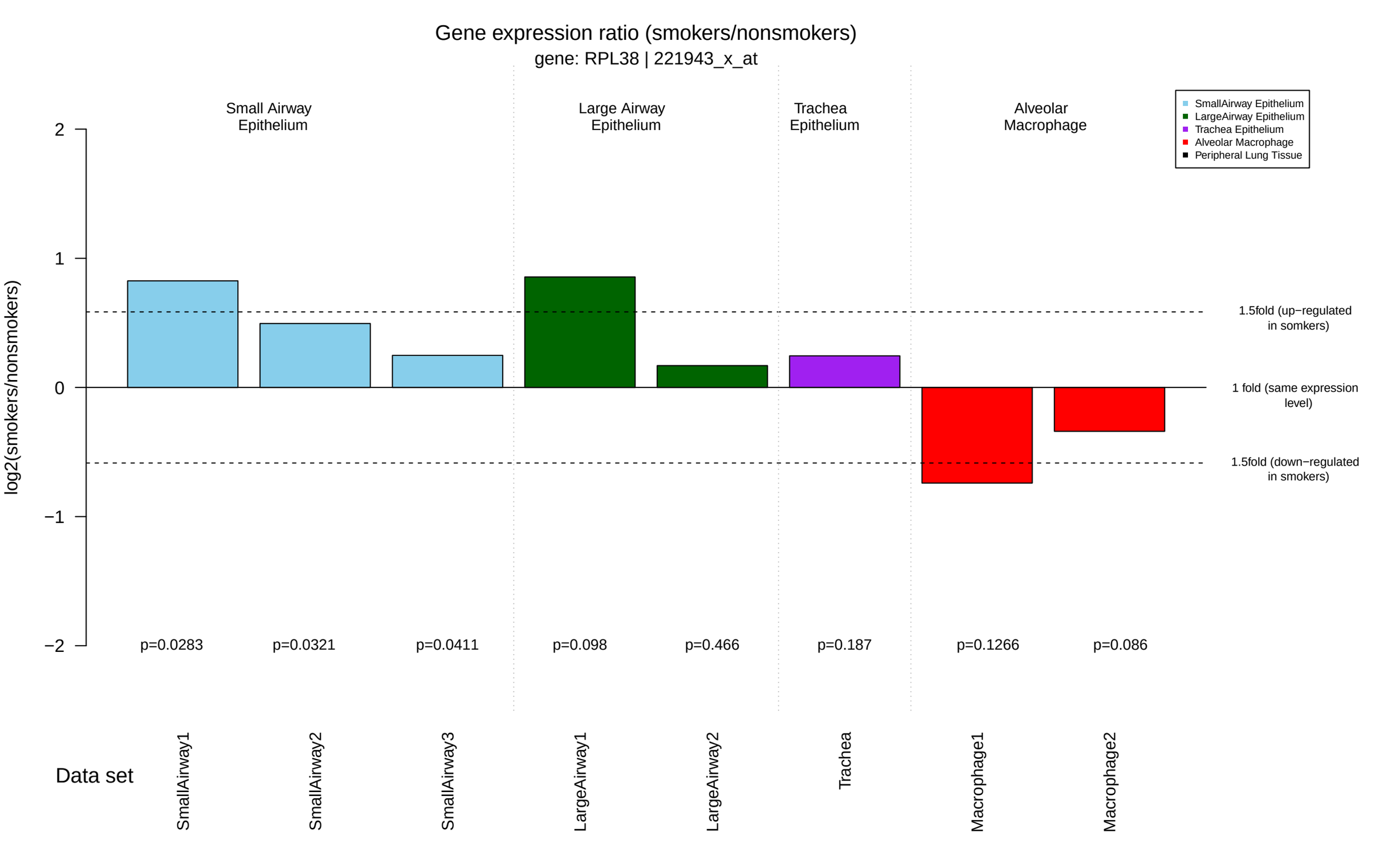
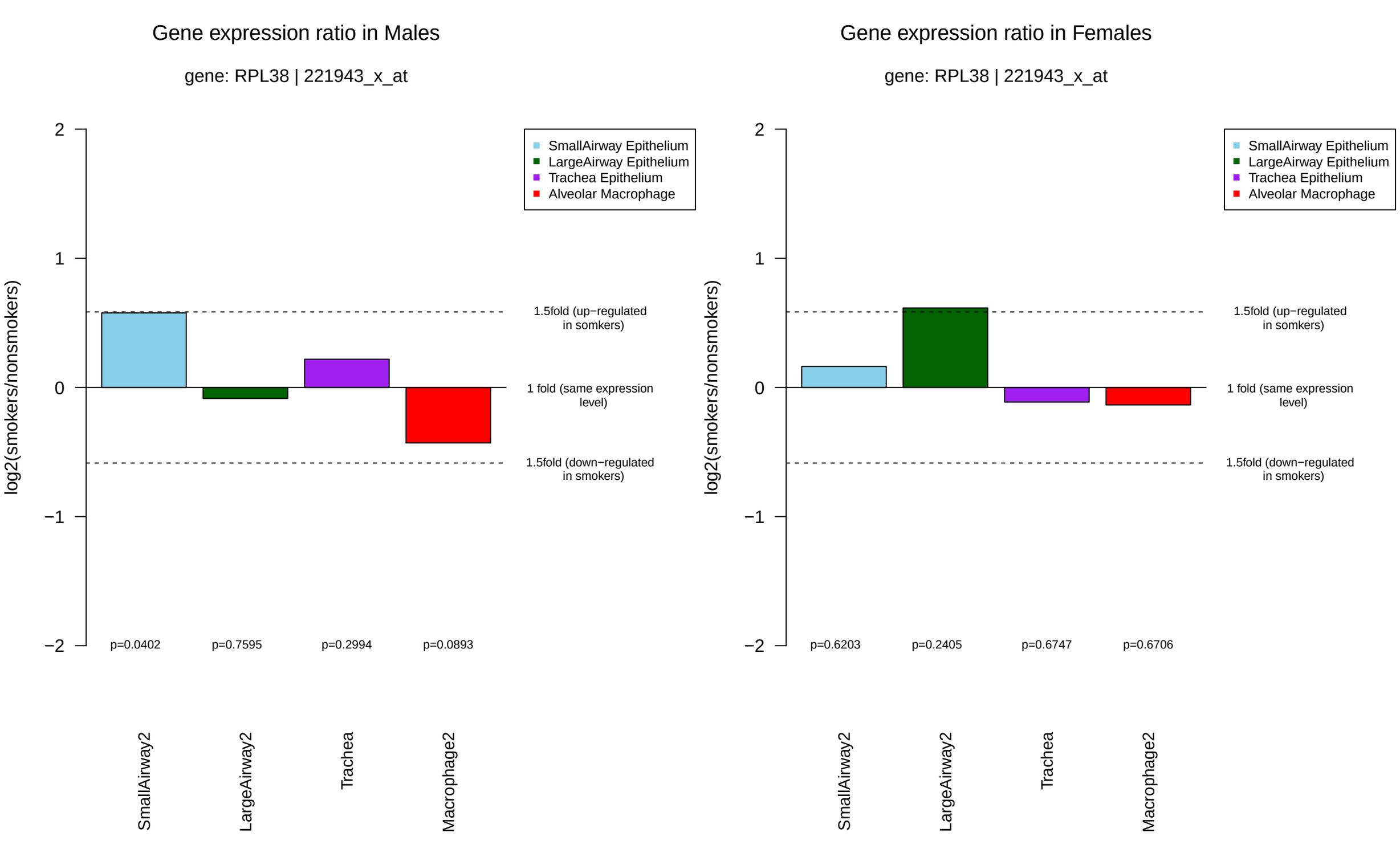
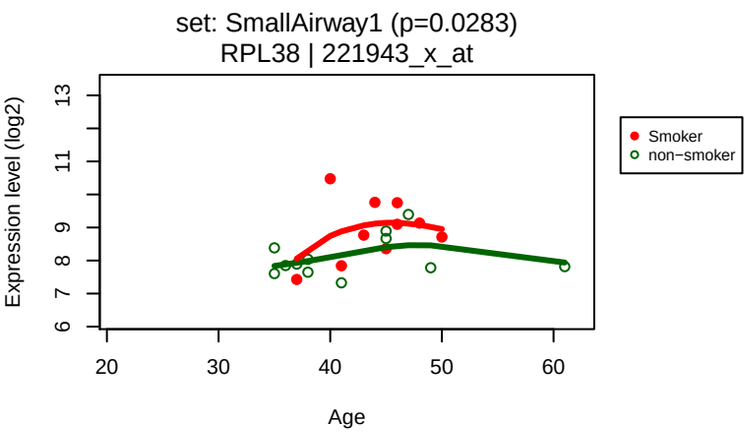
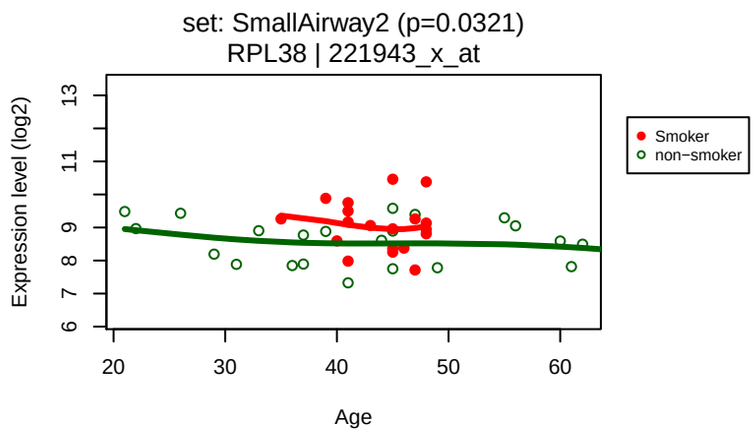
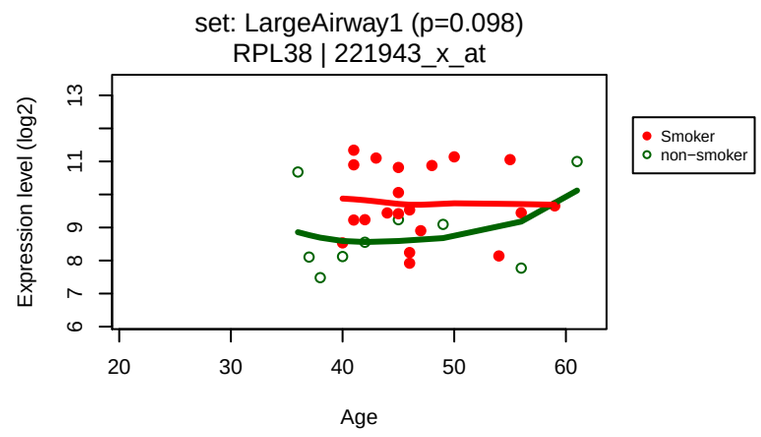

Gene expression of smokers and nonsmokers
gene: RPL38 | 221943_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0283
p=0.0321
p=0.0411
p=0.098
p=0.466
p=0.187
p=0.1266
p=0.086
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

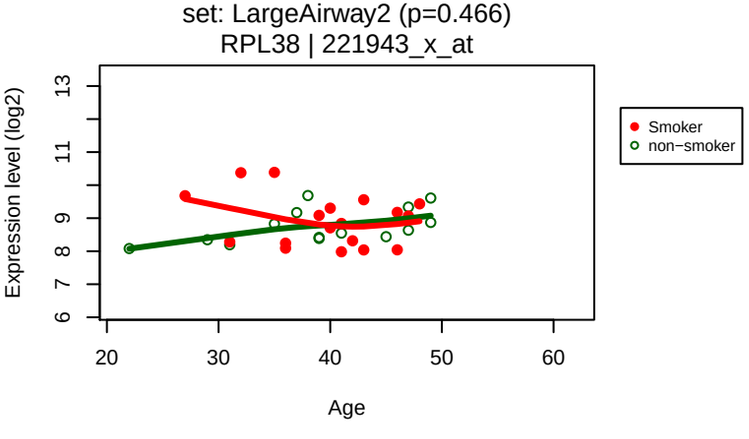

Gene expression ratio (smokers/nonsmokers)
gene: RPL38 | 221943_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0283
p=0.0321
p=0.0411
p=0.098
p=0.466
p=0.187
p=0.1266
p=0.086
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

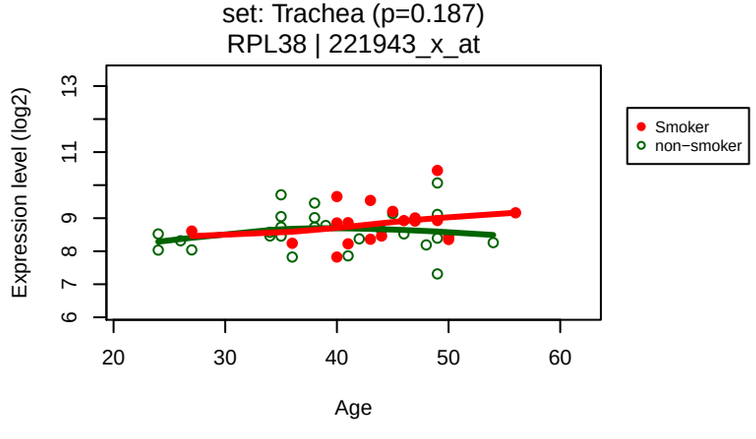

Gene expression ratio in Males
gene: RPL38 | 221943_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0402
p=0.7595
p=0.2994
p=0.0893
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL38 | 221943_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6203
p=0.2405
p=0.6747
p=0.6706
SmallAirway2
LargeAirway2
Trachea
Macrophage2

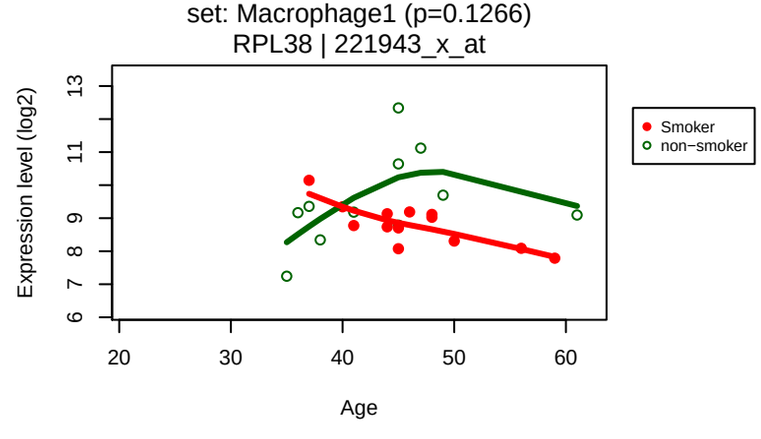

set: SmallAirway1 (p=0.0283)
RPL38 | 221943_x_at
Expression level (log2)
Age
Smoker
non−smoker

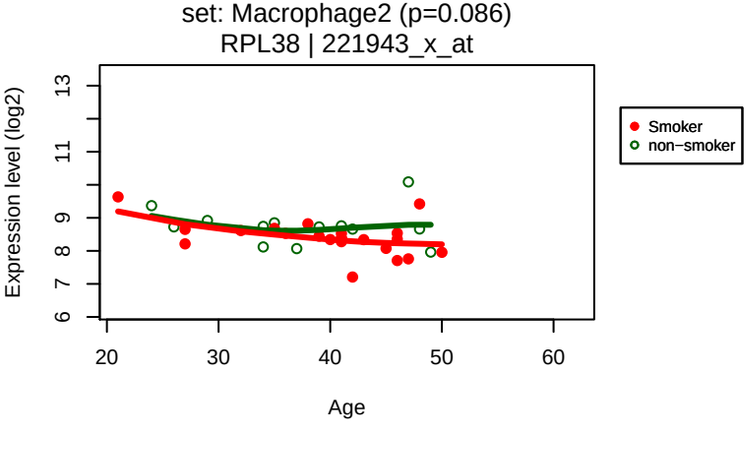

set: SmallAirway2 (p=0.0321)
RPL38 | 221943_x_at
Expression level (log2)
Age
Smoker
non−smoker

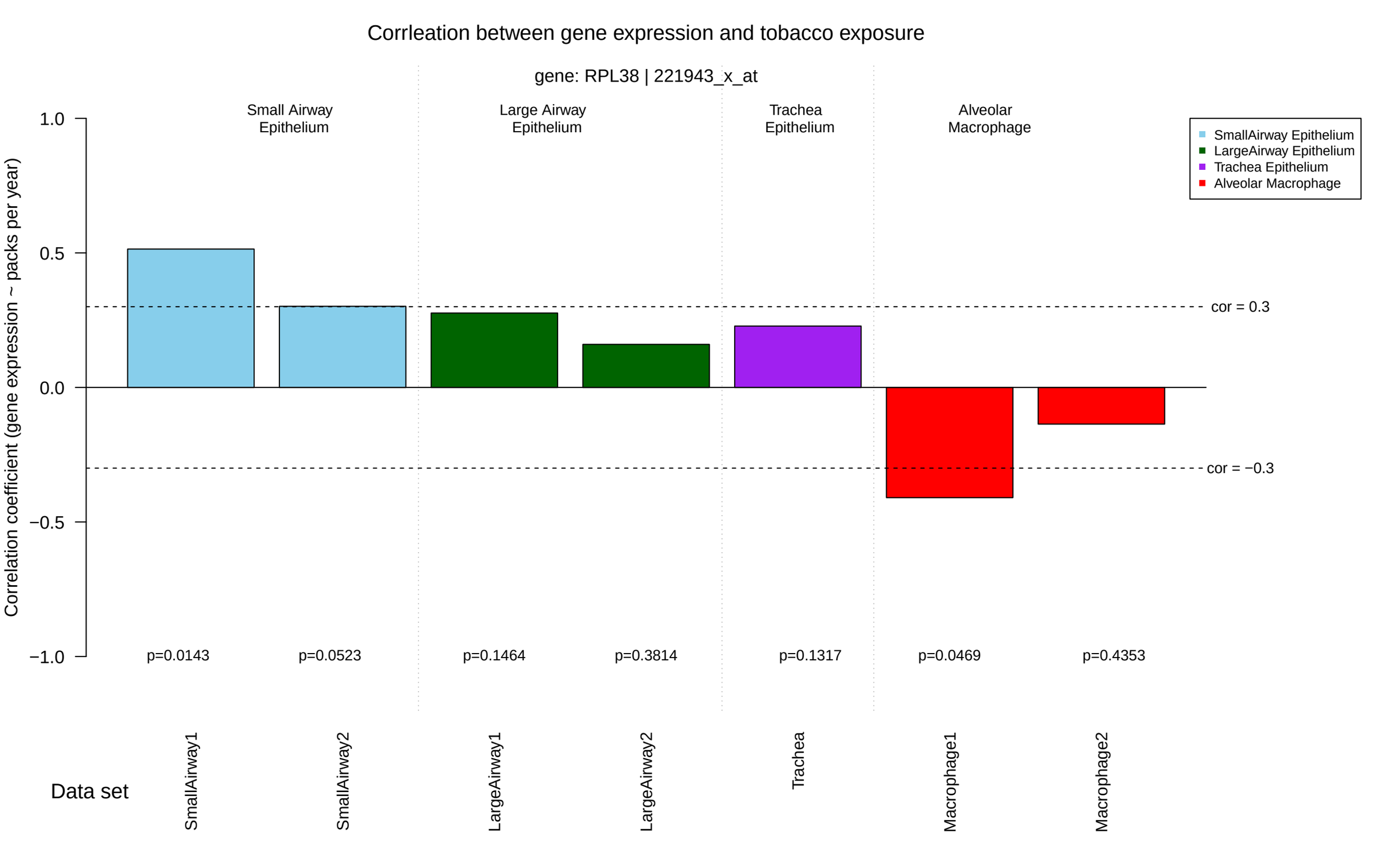

set: LargeAirway1 (p=0.098)
RPL38 | 221943_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.466)
RPL38 | 221943_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.187)
RPL38 | 221943_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1266)
RPL38 | 221943_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.086)
RPL38 | 221943_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL38 | 221943_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0143
p=0.0523
p=0.1464
p=0.3814
p=0.1317
p=0.0469
p=0.4353
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2